

ShambaShield: Scaling climate resilience for smallholders



INITIATIVE ON
Climate Resilience

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26-11-2024



Contents

- Overview 2**
- A package of solutions..... 3**
- Why invest in ShambaShield 5**
- Impact at scale 7**



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Overview

ShambaShield is a comprehensive financial and climate innovation bundle designed to transform the resilience and productivity of smallholder farmers in Kenya. By integrating climate-smart advisories, tailored financial products, and risk mitigation tools, ShambaShield addresses the core barriers farmers face in accessing resources, enhancing productivity, and adapting to the growing impacts of climate change. With a focus on inclusivity, particularly targeting women and youth, ShambaShield aims to make climate-smart farming and financial inclusion a reality for millions of farmers.

Our vision

“To scale ShambaShield across Kenya and East Africa, reaching over 500,000 smallholder farmers, at least 35% of whom are women and 15% youth. The goal is to increase financial inclusion, agricultural resilience, and ultimately improve livelihoods in climate-vulnerable regions.

A package of solutions

1. **Climate-Smart Credit Scoring Tool:** A unique tool that integrates climate data into credit risk assessments, providing a fair evaluation of a farmer's creditworthiness while accounting for climate risk. This reduces the perception of risk for lenders and increases farmers' access to finance.
2. **Digital Agro-Advisories and literacy through edutainment:** A popular TV and radio show reaching over 2 million farmers in Kenya and Zambia, delivering climate and financial literacy in an accessible format. By leveraging mass media, SSU empowers farmers with the knowledge to adopt climate-smart practices and make informed financial decisions.
3. **Inclusive Financial Products:** ShambaShield collaborates with microfinance institutions (MFIs) and agribusinesses to provide solutions that link credit, insurance, and technical assistance. Special focus is placed on group-based lending to help women access loans without traditional collateral.
4. **Insurance-Linked Credit:** Provides a financial safety net by linking credit to climate insurance, ensuring that smallholder farmers can repay loans even during adverse climatic events. This makes it easier for farmers, especially women and youth without collateral, to access funding for climate-adaptive investments.

Integrated approach

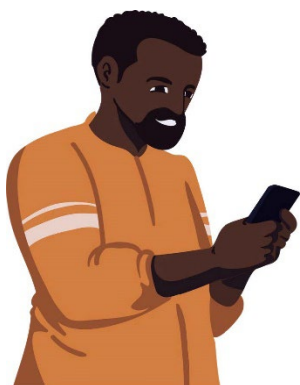
ShambaShield is not merely a bundled product but a system of interconnected solutions. Financial literacy feeds into the credit scoring process, helping farmers understand how to improve their creditworthiness. Credit is linked to insurance, ensuring that farmers are protected from climate-related risks. This creates a feedback loop where farmers are empowered to make informed decisions, access credit, and protect their investments.

The MVP for ShambaShield is simple integration of climate-smart advisories and literacy components with insurance and/or credit products—focusing on climate resilience. This simple yet transformative bundle allows smallholder farmers to access credit, secure insurance against climate risks, and receive the information needed to enhance productivity.



Goals, benefits, and capabilities of ShambaShield in Kenya and beyond

Scaling Element	Objective	Example
Goals	Adapting ShambaShield to address multiple types of climate risks and financial needs.	Providing crop insurance alongside credit for maize farmers to cover climate risks such as drought and pests.
Scaling Behaviours of Users	Focusing on changes in how farmers use ShambaShield, encouraging proactive climate adaptation.	A behavioural campaign encouraging users to access credit before droughts instead of during crises and adopt early land preparation.
Benefits Generated	Focusing on the increased positive outcomes on farmer income and climate resilience.	35% increase in farm income reported by participating farmers; 25% reduction in loan defaults through insured credit models.
Used Resource Base	Considering the increase in investment, partnerships, and infrastructure for scaling.	Collaboration with key partners to leverage digital services, enhancing reach to over 700,000 rural users.
Capabilities to Use	Focusing on improving farmer financial literacy and technical adoption of tools.	Conducting training programs for 100,000 farmers to boost their understanding of climate risks and their credit options.



Why Invest in ShambaShield?

ShambaShield is addressing a critical gap in agricultural finance and climate resilience for smallholder farmers. Investing in ShambaShield means supporting a solution that is grounded in a deep understanding of the market needs, designed for scalability, and backed by evidence of impact.

The USP of ShambaShield lies in its interconnected approach—combining financial literacy, credit, and insurance to create a comprehensive support system that empowers farmers. Unlike traditional financial products, ShambaShield directly addresses the vulnerabilities of smallholders by integrating financial inclusion with climate resilience, creating a unique feedback loop that amplifies impact. With proven scalability and a focus on inclusivity, ShambaShield stands out as a game-changer for millions of farmers in Kenya and beyond.

Proven market need

Smallholder farmers face significant barriers to accessing finance, particularly due to climate risks and a lack of financial history. ShambaShield directly addresses these challenges by providing tailored financial solutions and linking them to advisory services, creating a comprehensive approach to resilience. Over 40% of smallholder farmers in Kenya cite climate risk as their biggest barrier to accessing credit.

Market demand

The increasing demand for climate-resilient solutions in agriculture presents a unique market opportunity. Global market studies indicate that the demand for climate-smart agricultural solutions is expected to grow by over 20% annually by 2030. ShambaShield's innovation package meets the critical need for financial inclusion, particularly among underserved groups such as women and youth, while contributing to climate resilience. Currently, over 60% of smallholder farmers in Kenya lack access to appropriate financial services, and this gap presents a significant market need that ShambaShield is addressing. This positions ShambaShield as an impactful and scalable solution in the agricultural finance market.

Compelling business case

By integrating climate-smart advisories, credit, and insurance, ShambaShield presents a sustainable financial model that de-risks investments for financial institutions. Insurance-backed credit significantly lowers the risk of non-performing loans, thereby creating a favorable environment for investment. Additionally, group-based lending models and partnerships with MFIs create a solid base for scalability and return on investment. The bundled nature of these solutions increases the efficiency and effectiveness of financial support, reducing transaction costs and enhancing uptake among farmers. In Kenya, MFIs have seen a 25% reduction in default rates when insurance products are linked to credit (source: Kenya Microfinance Impact Report, 2023).

Proven impact

The Shamba Shape Up TV program has demonstrated significant improvements in the adoption of climate-smart practices among smallholder farmers, reaching millions of households. Insurance-linked credit has helped vulnerable farmers access credit for the first time, improving both their livelihoods and financial resilience. Studies show that 60% of SSU viewers have adopted at least one new climate-smart practice.

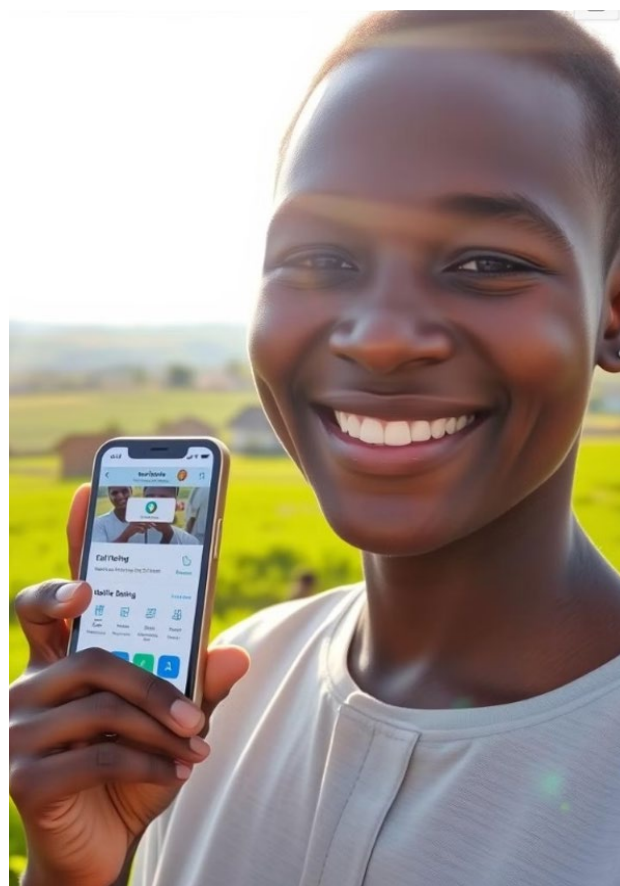
Scalable model

ShambaShield leverages mass communication, digital tools, and partnerships to ensure scalability. The package is designed for phased implementation, starting with pilot regions to refine the approach before expanding to broader areas across Kenya and East Africa. By 2025, ShambaShield aims to reach over 500,000 farmers. Working with key partners for impact at scale like Kenya Meteorological Department, the Mediae, DigiFarm and other partners, ShambaShield ensures that financial services, agro-advisories, and insurance products reach even the most vulnerable farmers.

Possible investments

Aspect	Development	Early Delivery	Scaling	Impact
Funded by	Grants from international donors	Catalytic funds from public/private sector	Grants from international donors	Private sector, civil society, government stakeholders
Led by	Researchers, agricultural experts.	Innovators and financial institutions.	Government agencies, large agribusinesses.	All stakeholders, including NGOs and communities.
Implemented by	ShambaShield research and design teams.	Initial pilot projects, MFIs.	Full-scale public-private partnerships.	Integration into government policies, large-scale uptake.
Focus On	Innovation readiness and tool development.	Validated use cases.	Complementary solution delivery.	Benefits realized at the ecosystem level.
Aim At	Technical performance and user needs.	Initial market uptake.	Adoption and replication.	Sustainable impact on agricultural resilience.
Build Capacity Of	Researchers, local MFIs.	Extension agents, technicians.	Farmers, cooperatives, institutions.	National agricultural networks.
How Do We Know It's Done Well?	Scientific studies, evaluation reports.	Achievement of pilot success targets.	Increased adoption without direct project support.	Integration of climate-adaptation practices. Increase in financial literacy and access to finance.

Note: Current innovation stage is highlighted in green



Impact at scale

ShambaShield follows a collaborative scaling pathway involving government agencies, private sector partners, and local community organizations. The aim is to establish a financially sustainable model that combines donor funding, private investment, and revenue from financial products. The focus is on reducing dependency on grants by making ShambaShield's offerings self-sustaining in the long term.

- **Partnership Expansion:** Engaging with additional MFIs and financial institutions to ensure broader adoption and access to the Climate-Smart Credit Scoring Tool. Partnerships with key institutions will support capacity building at a massive scale.
- **Capacity Building:** Training programs for loan officers and farmers are crucial. Modules will focus on digital literacy, use of the credit scoring tool, and climate risk understanding.
- **Platform Improvement:** The Climate-Credit Scoring System will be enhanced using human-centric design principles to ensure ease of use by MFIs and farmers. Improvements in data integration, user experience, and compatibility with existing systems are key priorities.
- **Community Awareness Campaigns:** Launching awareness campaigns using channels like radio, SMS, social media, and edutainment platforms such as Shamba Shape Up to improve farmer awareness and trust in financial products.

Scaling ShambaShield: Possible dimensions of scaling

Dimension	Objective	Example
Users	Expanding ShambaShield's reach to a larger number of smallholder farmers, focusing on underserved communities.	From 500,000 farmers in Kenya to 1 million farmers across East Africa, focusing on rural regions without financial inclusion.
Use Cases	Expanding the potential applications of ShambaShield to cover different crops and regions.	Using the Climate-Smart Credit Scoring Tool initially for maize and later for cash crops like coffee and tea in various regions.
Times	Increasing the frequency of engagement and repeat use of ShambaShield products and services.	Agro-advisory messages delivered twice weekly instead of monthly, increasing the likelihood of adoption of suggested practices.
Types/Versions	Diversifying features, tools, and services to meet different farming contexts.	Adding localized advisories and climate risk-specific credit products for pastoralists and horticulturists in different climatic zones.
Geographies	Expanding ShambaShield's presence across new regions.	Expanding from Kenya to include Ethiopia, Zambia, and Uganda, targeting drought-prone regions with high climate risk.

Join Us in Scaling Climate Resilience

ShambaShield is an initiative to promote financial inclusivity and climate resilience for vulnerable communities. We invite donors, private investors, and partners to support scaling this effort, extending financial security and climate adaptation to smallholder farmers who are most affected by climate change.

For more information on how to partner or invest in ShambaShield, please contact **Shalika Vyas, Research Specialist** at s.vyas@cgiar.org.

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